



RS6240

Hardware Design and PCB Layout

Guidelines

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1 Feature

- **FMCW mmWave Radar**
 - 57-66GHz in coverage, 8GHz scanning bandwidth
 - 2 transmitting channels and 4 receiving channels, equipped with AiP packaged antennas
 - Integrated RFPLL, DPLL, Transmitter, Receiver, Analog-to-Digital Conversion, and Baseband
 - HWA is used for 1D/2D FFT, static clutter removal, and CFAR operations
- **2.4GHz Wireless Connection**
 - Protocol: Supports BLE5.3, BLE MESH, IEEE802.15.4, Thread1.3, Matter1.1, and 2.4GHz proprietary protocols
 - Bluetooth PHY: 1Mbps, 2Mbps, Long-range S2 (500Kbps), S8 (125Kbps)
- **Application System**
 - Dual-core 32-bit RISC-V CPU, featuring FPU (up to 208MHz) and I/D cache, for radar post-processing, wireless, and applications
- **Host and Device Interfaces**
 - x3 UART
 - x2 QSPI with XIP
 - X2 I2C
 - x8 PWM
 - x8 GPADC
 - x1 LIN
 - x1 CAN-FD
- **Clock Source**
 - 32/40MHz crystal oscillator as clock reference
 - Internal LSI and MSI RC oscillators
 - Optional 32KHz crystal for low-power operation
- **Power Management**
 - Single power supply ranging from 1.7V to 5.5V
 - Built-in LDO network for enhancing PSRR
 - BOM optimization and power consumption optimization modes
- **Ultra-low Power Consumption Characteristic**
 - Power status: Off, Standby, Idle, and Active (Sensing, Processing, and Connecting)
- **Encapsulation**
 - AiP FCCSP: 12.4 x 8.8mm² with 159 pins
- **Temperature Range**
 - Operating junction temperature: -40°C to 125°C

2 Pin Description

2.1 Chip Pin Diagram

	1	2	3	4	5	6	7	8	9	10	11	12	13
A	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	ANT	GND
B	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND
C	GND	GND	GND	GND	GND	GND	GND	GND	XL1	XL2	GND	GND	GND
D	GND	GND	V13T	VRF	GND	GND	GND	GND	GND	GND	GND	GND	GND
E	GND	GND	V13R	VRF	GND	GND	GND	V18A	V18C	V13M		PA15	PA14
F	GND	GND										PA13	PA12
G	GND	GND										PA11	PA10
H	GND	GND										PA9	PA8
J	GND	GND										PA6	PA7
K	GND	GND										PA5	GND
L	GND	GND										PA4	PA3
M	GND	GND										PA2	PA1
N	GND	GND										NRST	PA0
P	GND	GND	GND	GND	GND	GND	GND		GND	VDIG	VINT	VEXT	VDDIO
R	GND	GND	GND	GND	GND	GND	GND		GND	GND	VINT	VCC	VCC
T	GND	GND	GND	GND	GND	PB3	GND	GND	GND	PB4	VDIG	GND	GND
U	GND	GND	GND	GND	GND	PB2	PB7	PB6	PB5	PB1	PB0	GND	GND
V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND

Figure 2-1 Chip Pin Diagram

2.2 Pin Description

Table 2-1 Pin Description

Pin Name	I/O	Pin No.	Pin Description
VCC	Power In	R12,R13	Power input, 1.7~5.5V
VINT	Power Out	P11,R11	INT LDO output
VEXT	Power Out	P12	EXT LDO output
VDIG	Power Out	P10	DIG LDO output
VDDIO	Power In	P13	GPIO power input, 1.7~3.6V
V18A	Power In	E8	MMW analog power supply input, 1.5~2.5V
V18C	Power In	E9	MMW analog power supply input, 1.5~2.5V
V13T	Power In	D3	MMW RF transmitting power supply input, 1.2~2.5V

V13R	Power In	E3	MMW RF receiving power supply input, 1.2~2.5V
VRF	Power Out	D4,E4	MMW RF LDO output, default 1.0V
V13M	Power In	E10	2.4G analog power input, 1.2~2.5V
XL1	Analog	C9	32/40M crystal oscillator output
XL2	Analog	C10	32/40M crystal oscillator input
ANT	RF	A12	2.4G antenna port
GND	Ground	/	Ground Pin, Short directly to module ground plane
NRST	In	N12	Reset the chip, and default to pull up to VDDIO
PA0	In/Out	N13	GPIO input/output, Wakeup IO
PA1	In/Out	M13	GPIO input/output, Wakeup IO
PA2	In/Out	M12	GPIO input/output, Wakeup IO
PA3	In/Out	L13	GPIO input/output, Wakeup IO
PA4	In/Out	L12	GPIO input/output, Wakeup IO
PA5	In/Out	K12	GPIO input/output, Wakeup IO
PA6	In/Out	J12	GPIO input/output, Wakeup IO
PA7	In/Out	J13	GPIO input/output, Wakeup IO
PA8	In/Out	H13	GPIO input/output, Wakeup IO
PA9	In/Out	H12	GPIO input/output, Wakeup IO
PA10	In/Out	G13	GPIO input/output, Wakeup IO
PA11	In/Out	G12	GPIO input/output, Wakeup IO
PA12	In/Out	F13	GPIO input/output, Wakeup IO
PA13	In/Out	F12	GPIO input/output, Wakeup IO
PA14	In/Out	E13	GPIO input/output, Wakeup IO
PA15	In/Out	E12	GPIO input/output, Wakeup IO
PB0	In/Out	U12	GPIO input/output, Wakeup IO
PB1	In/Out	U11	GPIO input/output, Wakeup IO
PB2	In/Out	U7	GPIO input/output
PB3	In/Out	T6	GPIO input/output
PB4	In/Out	T10	GPIO input/output
PB5	In/Out	U10	GPIO input/output
PB6	In/Out	U9	GPIO input/output
PB7	In/Out	U8	GPIO input/output

2.3 GPIO Multiplexing Table

Table 2-2 GPIO Multiplexing Table

GPIO Name	Func2	Func3	Func4	Func5	Func6	Func7	Func after rst
PA0/WIO0	SPI0_CS	UART1_TX	CANFD_TX		PWM0		OFF/No-Pull
PA1/WIO1	SPI0_CLK	UART1_RX	CANFD_RX		PWM1		OFF/No-Pull
PA2/WIO2	SPI0_MOSI	UART1_RTS	I2C0_SDA	UART2_TX /LIN_TX	PWM2		OFF/No-Pull
PA3/WIO3	SPI0_MISO	UART1_CTS	I2C0_SCL	UART2_RX /LIN_RX	PWM3		OFF/No-Pull
PA4/WIO4	SPI0_HOLD	I2C0_SDA			PWM4		OFF/No-Pull
PA5/WIO5	SPI0_WP	I2C0_SCL			PWM5		OFF/No-Pull
PA6/WIO6	UART1_RTS	UART2_TX	I2C1_SDA		PWM6	GPADC6	OFF/No-Pull
PA7/WIO7	UART1_CTS	UART2_RX	I2C1_SCL	PSRAM_CS	PWM7	GPADC7	OFF/No-Pull
PA8/WIO8/LXL1 ¹	UART1_TX	I2C1_SCL			PWM0	GPADC0	OFF/No-Pull
PA9/WIO9/LXL2 ¹	UART1_RX	I2C1_SDA			PWM1	GPADC1	OFF/No-Pull
PA10/WIO10	I2C1_SDA	CANFD_TX			PWM2	GPADC2	OFF/No-Pull
PA11/WIO11	I2C1_SCL	CANFD_RX			PWM3	GPADC3	OFF/No-Pull
PA12/WIO12	UART2_RX	UART1_RTS			PWM4	GPADC4	OFF/No-Pull
PA13/WIO13/TEST ²	UART2_TX	MCO_CLK			PWM5	GPADC5	IN/UP
PA14/WIO14	TWIO_SDA	UART2_RX		CANFD_TX	PWM6		OFF/No-Pull
PA15/WIO15	TWIO_SCL	UART2_TX		CANFD_RX	PWM7		OFF/No-Pull
PB0/WIO16	UART0_TX				PWM0		OFF/UP
PB1/WIO17	UART0_RX				PWM1		OFF/UP
PB2	UART0_RTS	FC_WP	SPI0_WP		PWM2		OFF/UP
PB3	UART0_CTS	FC_HOLD	SPI0_HOLD		PWM3		OFF/UP
PB4	FC_CS	UART2_RX	SPI0_CS		PWM4		OFF/No-Pull
PB5	FC_CLK	UART2_TX	SPI0_CLK		PWM5		OFF/No-Pull
PB6	FC_MOSI	I2C0_SDA	SPI0_MOSI		PWM6		OFF/No-Pull
PB7	FC_MISO	I2C0_SCL	SPI0_MISO		PWM7		OFF/No-Pull

- (1) LXL1 and LXL2 are 32K crystal oscillator pins that are multiplexed with PA8 and PA9. When used for analog functions, the GPIO function needs to be disabled.
- (2) When the chip is powered on for the first time or restarted after a reset, PA13 will be configured in input with pull-up mode. If this IO is pulled to GND at this time, the chip will enter ATE test mode. Therefore, PA13 should be avoided being pulled low by default.
- (3) The chip defaults to using UART0 of PB0 and PB1 as the default download and control serial port.

3 Peripheral Interface Characteristics

3.1 GPIO

RS6240 provides 18 GPIO, each with multiple multiplexing functions.

GPIO configuration features:

- Three modes: Input, Output, and High Impedance
- Input configurable pull-up and pull-down
- The output drive capability is set to 20mA by default, with one setting at 20mA and a maximum configuration of 80mA
- IO output supports both push-pull mode and open-drain mode

Note:

- In low-power scenarios, all GPIO support triggering the system wake-up through interrupt mode.
- GPIO PA13 has the function of enabling the chip to enter test mode. Before the chip's NRST reset signal is released, PA13 will be in a pull-up state. When designing the peripheral circuit, it is necessary to ensure that PA13 is at a high level or floating when the chip is started.

3.2 UART

RS6240 supports UART0, UART1, and UART2, among which UART0 and UART1 feature hardware flow control. UART is compatible with the 16550 industry standard and supports baud rates up to 3Mbps.

UART possesses the following characteristics:

- Programmable transmit and receive baud rate
- Baud rate self-checking function
- Supports 64-byte receive and transmit FIFO
- Supports data lengths of 5/6/7/8 bits
- Support 1/1.5/2 stop bits
- Support parity check
- Supports DMA high-speed data communication
- Supports UART wake-up function, with UART RX having the ability to detect rising edge changes and trigger interrupts. The number of rising edges detected can be configured from 1 to 8. In low-power scenarios, the system is woken up after UART RX receives a matching rising edge.

3.3 I2C

RS6240 supports I2C0 and I2C1 interface communication and features a rich Pinmux.

I2C possesses the following characteristics:

- Can be configured as master and slave modes
- Supports standard mode (100Kbit/s)
- Supports high-speed mode (400Kbit/s)
- Supports high-speed mode enhancement (1000Kbit/s)
- Supports both 7-bit addressing mode and 10-bit addressing mode
- Supports receive and transmit FIFO depths of 8 bytes
- Supports I2C wake-up function. In low-power scenarios, I2C supports 7-bit addressing matching to trigger the wake-up system.

3.4 PWM

The chip provides four PWM generators, each of which outputs two PWM signals and a pair of complementary signals.

The PWM characteristics are as follows:

- 8-channel PWM output and 4-channel capture input (PWM0, PWM2, PWM4, and PWM6)
- Configurable dead zone
- Supports pulse signal, periodic signal, and complementary signal output
- Adjustable duty cycle from 0-100%

3.5 LIN

The LIN controller supports the following features:

- Supports transmission at rates up to 1mHz
- Support the host mode to initiate the sending and receiving of messages with different frame IDs
- Support the processing of received and sent messages with different IDs in slave mode response
- Support arbitration adaptation under multiple slaves
- Adjustable length of different separation fields
- Supports two verification modes: traditional verification and enhanced verification
- The host mode can be set with different frame slot widths

3.6 CAN-FD

The CAN controller supports the following features:

- CAN2.0B: Supports payloads of up to 8 bytes, with a data rate of up to 1 Mbit/s
- CAN-FD: It supports payloads of up to 64 bytes, with the data rate being limited by the transceiver and the CAN-CTRL core clock frequency
- TTCAN: Supports time-triggered CAN and can be used in conjunction with CAN 2.0B or CAN FD
- Configurable receive buffer size: Three buffer slots, with FIFO (First-In First-Out) behavior
- Two sending buffers: one primary sending buffer and three secondary sending buffers
- Four acceptance filters: The acceptance code can be checked through bits
- Extended functions: Single transmission mode; Listening-only mode

3.7 SPI

RS6240 supports a quad-serial SPI interface (QSPI). The QSPI protocol supports half-duplex, full-duplex, synchronous, and serial communication with peripherals. This interface can be configured in either master or slave mode. There are four possible combinations of serial clock phase and polarity.

The following features are supported:

- The clock frequency can reach up to 96 MHz
- Supports Motorola SPI mode
- Programmable master mode or slave mode
- Supports standard SPI, dual SPI, and quad SPI modes
- Programmable data width, ranging from 4 to 32 bits
- The FIFO depth for both sending and receiving data is 32 words
- Programmable sampling delay time in main mode
- Programmable Serial Clock Polarity (SCPOL) and Serial Clock Phase (SCPH)

- Supports interrupt and DMA
- Programmable instructions, address length, and wait cycles in main mode
- Support clock detection to wake up the system from standby state

4 Peripheral Interfaces Usage Guide

4.1 Precautions for GPIO Usage

Certain GPIO possess special characteristics, and the following points should be noted in application scenarios:

- (1) PA0~PA15 and PB0~PB1 have WIO function, supporting the triggering of system wake-up through power consumption scenario interrupt. Other GPIO do not have this function.
- (2) PA13 has the function of entering the chip ATE test mode (this function is only used for chip production testing and is not applicable to scheme development). When the chip NRST is pulled low or powered on for the first time, PA13 is in a weak pull-up state. When NRST is pulled high to release or when powered on for the first time, PA13 cannot be pulled low. After the chip starts up, PA13 can be used normally.
- (3) When the chip is powered on or reset to start, PB0 and PB2 output high voltage levels, which are maintained for tens of microseconds.
- (4) After the chip is powered on or reset and starts up, PA0 and PA1 are in a weak pull-up input state.
- (5) For unused IO, leave them unconnected.

4.2 Recommended Interface Usage

4.2.1 UART Interface Usage Recommendations

- 1) PB0 and PB1 are the default UART0, serving as the chip programming firmware interface and communication interface, and supporting low-power wake-up function;
- 2) PB2~PB7 are dedicated external Flash interfaces for applications;
- 3) PA0 is the default UART1-TX reporting data interface in firmware. In application design, PA0 is preferably considered as a debug or reporting data interface.
- 4) The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.
- 5) To facilitate radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.

4.2.2 IIC Interface Usage Recommendations

- 1) PA4 and PA5 are the default IIC0, serving as the chip programming firmware interface and communication interface, and supporting low-power wake-up function;
- 2) PB2~PB7 are dedicated external Flash interfaces for applications;
- 3) PB0 is the trigger signal for the chip to enter forced programming mode.
- 4) The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.
- 5) To facilitate radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.

4.2.3 SPI Interface Usage Recommendations

- 1) PA0~PA3 are the default SPI0, serving as the chip programming firmware interface and communication interface, supporting low-power wake-up function;
- 2) PB2~PB7 are dedicated external Flash interfaces for applications;
- 3) PB0 is the trigger signal for the chip to enter forced programming mode.
- 4) The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.

- 5) To facilitate radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.
- 6) **Note: If the SPI operating frequency is above 30MHz, please ensure that the SPI controller on the host side has the ability to adjust data sampling delay to ensure correct reading of radar data at different frequencies.**

4.3 Enter Programming Mode

Based on the specific hardware configuration, users can enable the chip to enter the programming mode through the following two schemes.

Option 1: The main controller can control the NRST pin through IO

- (1) Keep VCC powered on, with PB0 and PB2 at high voltage level;
- (2) Set NRST to low level and delay for 20ms;
- (3) Set NRST to high level and delay for 50ms;
- (4) Set PB0 (or PB2) to low level, and set NRST to low level;
- (5) Delay for 20ms, set NRST to high level;
- (6) Delay for 50ms, set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered programming mode.

Option 2: Suitable for systems with an external control power supply interruption mechanism

- (1) Power off our chips, Flash, or modules. If the external capacitor is 10uF, it is recommended to maintain the power-off state for 10 seconds. Please adjust the power-off duration based on the external capacitor and the actual power-down speed;
- (2) Set PB0 (or PB2) to low level;
- (3) Power on our chip and Flash or module again, delay for 50ms, and set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered programming mode.

4.4 Programming Interface

The chip provides three programming interfaces for firmware, namely UART, IIC, and SPI, and employs a polling method to configure the communication interface. The default programming interface configuration is as follows:

- (1) UART uses the UART0 interface of PB0 and PB1
- (2) IIC uses the IIC0 interface of PA4 and PA5
- (3) SPI uses the serial simplex mode of PA0~PA3

After the chip enters the forced programming mode, the UART, IIC, and SPI interfaces perform polling and handshake communication. Once the communication is successful, the firmware is downloaded and programmed.

5 Power-up Sequence

5.1 Power-up Sequence

This chip has no special requirements for the power-on sequence of various power supplies, but there are still the following considerations for different configuration schemes:

- (1) If SPI NOR Flash is used as the boot device, and the NRST signal is pulled up to VDDIO by default, it is not controlled by external signals:**

After the chip is powered on and NRST is released (voltage transitions to high level), it will immediately start and attempt to read the firmware from the SPI NOR Flash. On the other hand, the SPI NOR Flash needs to be powered up for a specified duration before it can be accessed. Generally, this chip solution uses VDDIO to power the SPI NOR Flash. Therefore, an RC delay circuit needs to be added around the NRST pin to ensure that during the power-on process, the SPI NOR Flash is powered up and undergoes the required delay before the chip's NRST signal is released.

Please ask the user to calculate the parameters of the RC delay circuit according to the power-on timing requirements specified in the SPI NOR Flash specification.

- (2) If SPI NOR Flash is used as the boot device and NRST is controlled by an external master, the chip can be reset and booted by manipulating NRST after power-on is completed and the external master is ready, ensuring that SPI NOR Flash has completed power-on.**

6 Packaging Instructions

6.1 Encapsulation Block Diagram

RS6240 adopts FCCSP packaging, with bump diameter of 350 μ m, pitch of 650 μ m, standoff of 280 μ m, dimensions of 12.4 x 8.8 x 0.81mm³, and 159 balls.

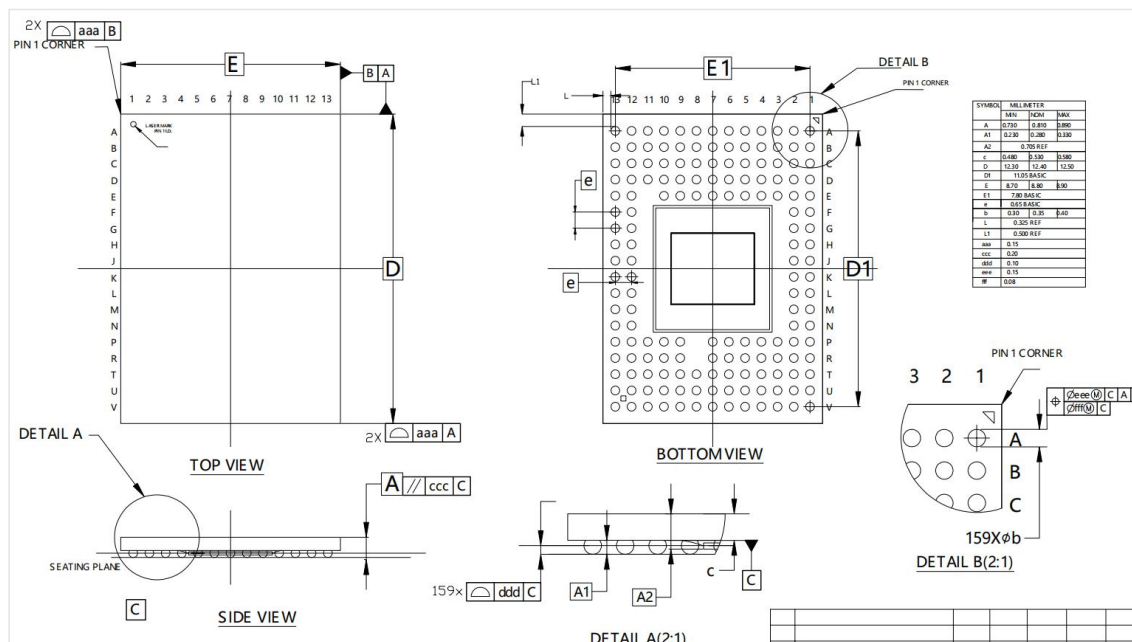


Figure 6-1 Packaging Block Diagram

6.2 PCB Pad Diagram

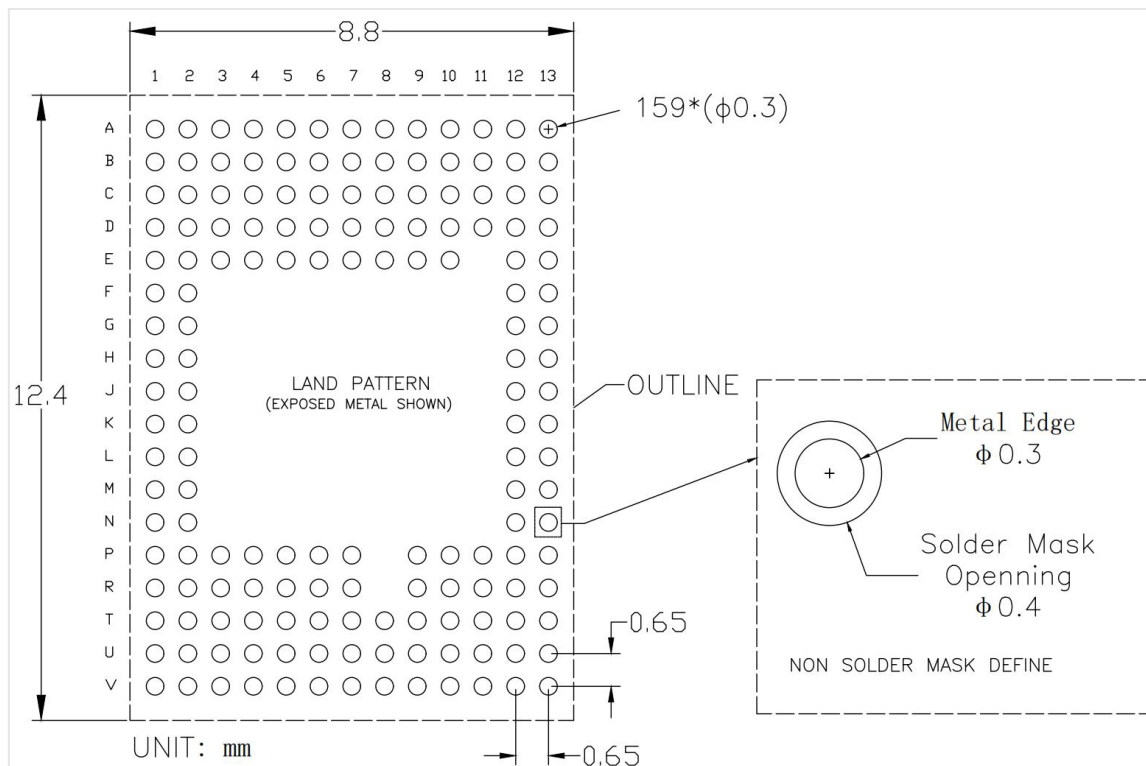


Figure 6-2 PCB Pad Diagram

7 Reference Schematic

7.1 Precautions

- **Power Supply**

- **OPTION A:** Compared to using the RS6240 internal LDO to supply power to V13, using an **external** DCDC for power supply will result in more power savings. If the user is not sensitive to power consumption, this part of the circuit can be removed, thereby saving BOM costs.
- VCC and VDDIO are powered externally. If VCC is lower than 3.6V, VCC and VDDIO can be tied together and powered by a single power supply.

- **Bluetooth**

- **OPTION B:** "32.768KHz" is used for the Bluetooth subsystem. If the user does not require Bluetooth functionality, this part of the circuit can be removed.
- **OPTION C:** "BLE ANTENNA" is used for the Bluetooth subsystem. If the user does not require Bluetooth functionality, this part of the circuit can be removed.

- **External Connection**

The main controller can control the radar through any of the interfaces UART0, I2C0, or SPI0, with UART0 being the default.

The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.

If there is a need for OTA firmware, PB0 can be pin-out to control the chip to enter the programming mode. For specific details, refer to Section 4.3.

For the convenience of radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.

- **OPTION D: CONNECT By I2C0**

- **OPTION E: CONNECT By SPI0**

- **PSRAM**

- **OPTION F:** "PSARM" is used for memory expansion. If users do not require functions such as attitude detection or high-performance human presence sensing, which involve large data volume operations, this part of the circuit can be removed. It is recommended to keep it.

- **For unused pins, leave them unconnected.**

7.2 Reference Schematic

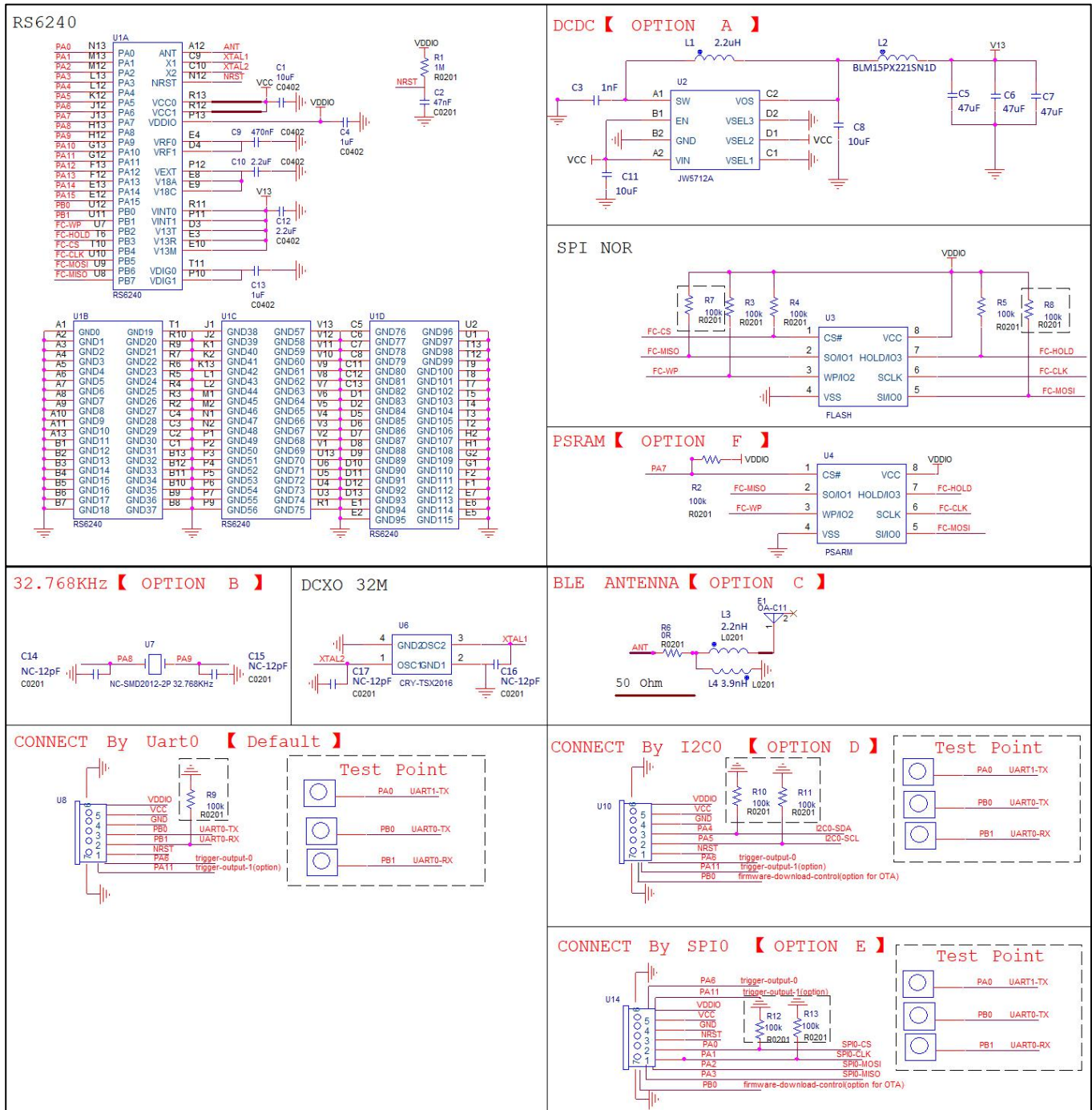


Figure 7-1 Reference Schematic

When designing the circuit, please carefully read the descriptions of each circuit module in the table below:

Table 7-1 Circuit Module Description

Module	Device	Description
RS6240	C1,C4,C9,C10,C12,C13	Chip decoupling capacitor (X5R, accuracy $\pm 10\%$, voltage resistance 6.3V), do not change the capacitance value.
	R1,C2	The NRST power-on RC delay ensures that the FLASH is ready before NRST is released. The parameters need to be set according to the power-on timing sequence of the flash. For details, refer to Section 5.1 Power-on Timing Sequence Description.
DCXO 32M	U6,C16,C17	The crystal specification is 32MHz, 12pF, with an accuracy of $\pm 10\text{ppm}$; C16 and C17 (COG, with an accuracy of $\pm 1\%$ and a voltage tolerance of 6.3V) are matching capacitors. The chip has built-in capacitors, so NC is the default setting.
32.768KHz [OPTION B]	U7,C14,C15	The low-frequency crystal oscillator operates at 32.768KHz with an accuracy of $\pm 20\text{ppm}$. C14 and C15 (COG, with an accuracy of $\pm 1\%$ and a voltage resistance of 6.3V) serve as matching capacitors and must not be left in a non-conducting (NC) state. Their capacitance values are determined based on the selected model. This crystal oscillator provides precise timing for the Bluetooth system in standby mode. If Bluetooth is not in use, it can be disabled.
BLE ANTENNA [OPTION C]	R6,L3,L4,E1	Bluetooth antenna and matching circuit; the matching circuit needs to be modified according to the selected antenna type, and the antenna feed signal trace needs to be impedance-matched to 50 ohms; if Bluetooth is not used, it can be deleted, and the signal can be left idle.
SPI NOR	R3,R4,R5,U3	R3, R4, R5, R7, and R8 are pull-up resistors (accuracy $\pm 5\%$); U3 is a flash memory, please select the type from the "Key BOM Selection Guide Manual". If U3 is a WINBOND flash memory, R7 and R8 need to be added for pull-up, so that the flash can enter the lowest sleep power consumption mode; when using flash memories from other brands, R7 and R8 can be removed.
PSRAM [OPTION F]	R2,U4	R2 is a pull-up resistor (accuracy $\pm 5\%$); U4 is a PSRAM, model APS1604M-3SQR-ZR, which shares SPI signals with flash and is used for time-sharing multiplexing; "PSARM" is used for memory expansion. If users do not need to implement functions such as attitude detection or high-performance human presence sensing, which require large amounts of data processing, this part of the circuit can be removed, but it is recommended to keep it and not mount it.
DCDC [OPTION A]	Selection description	Compared to using the internal LDO of RS6240 to supply power to V13, using an external DCDC for power supply will result in more power savings. If the user is not sensitive to power consumption, this part of the circuit can be removed, thereby saving BOM costs.
	L1,C3,C8,C11	JW5712A typical matching circuit L1: DCDC matching inductor, $\text{DCR} \leq 300\text{m}\Omega$, rated current $\geq 300\text{mA}$, such as LQM18PN2R2MFRL C: Matching capacitor (X5R, accuracy $\pm 10\%$, voltage resistance 6.3V)
	U2	JW5712A, a low-power DCDC, is not recommended for replacement with other models; if replacement with other models is necessary, they must meet the requirements of a quiescent current less than 600nA, an operating current greater than 400mA, and an output voltage of 1.3V. For more models, please refer to the "Key BOM Selection Guide Manual".
	L2	DCDC output filter bead, $1\text{K}\Omega@100\text{MHz}$, $\text{DCR} \leq 1.1\Omega$, rated current $\geq 250\text{mA}$, such as BLM15HG102SN1D.

	C5,C6,C7	DCDC output filter storage capacitor (X5R, accuracy $\pm 20\%$, voltage rating 6.3V), with varying capacitance values suitable for low-power applications with different power supply types: ➤ Batteries with high internal resistance, such as button batteries and dry batteries: Add a 200uF capacitor at the battery terminal, reserve 2 capacitor positions for standby, and attach 47uF capacitors to C5, C6, and C7. If the actual layout space is insufficient, the capacitance value can be appropriately reduced, and refer to "Section 9.2. DCDC Output Performance Measurement" to confirm the performance. ➤ Lithium batteries or batteries with low internal resistance such as capacitors: Reserve 2 capacitor positions at the battery terminal for spare use. Attach 10uF to C5 and C6, and leave C7 as NC.
CONNECT By UART0 [Default]		➤ The main controller can control the radar through UART0/I2C0/SPI0, with UART0 being the default. For ease of radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during PCB performance analysis.
CONNECT By I2C0 [OPTION D]		
CONNECT By SPI0 [OPTION E]		➤ VCC and VDDIO are powered externally. If VCC is lower than 3.6V, VCC and VDDIO can be tied together and powered by a single power supply. ➤ PA6/PA11 is typically used as an interrupt trigger signal to wake up the host. ➤ If there is a need for online firmware updates, PB0 needs to be pin-out to control the chip to enter programming mode. ➤ If VCC is not used as a power switch, the NRST signal needs to be connected to the main controller to assist in implementing abnormal reset or updating firmware online. ➤ For the pins RX (PB1) of UATR0, CS (PA0) and CLK (PA1) of SPI0, and SDA (PA4) and SCL (PA5) of I2C0, they cannot be in a floating state when the radar is in hibernation, otherwise the radar may experience abnormal leakage. If the main control terminal cannot set the corresponding signal pins to a fixed level during hibernation, external pull-down resistors R9/R10, R11/R12, and R13 need to be added to fix the level. If the main control terminal can set the level, they can be removed.

8 PCB Layout Guide

mmWave radar performance is sensitive to floor layout, spatial clearance, and component layout and routing. The following will provide design explanations for these three aspects.

8.1 Floor Layout

The radiation characteristics of mmWave radar vary with changes in the PCB board size. This is particularly evident in the E-plane radiation characteristics of mmWave radar. As the PCB size increases, the E-plane radiation pattern exhibits ripples and deteriorates at large angles. However, the H-plane radiation characteristics are less affected by the PCB board size and can be neglected. To eliminate the influence of the PCB board on the radar's radiation characteristics, the PCB board requires corresponding performance optimization design.

- Perform trapezoidal windowing in the E direction of the PCB (hollow out all layers of metal in the trapezoidal area), with $d=0.6\text{mm}$, $w\geq 3.75\text{mm}$, and $h\geq 3.75\text{mm}$.
- The surface layer of the blue-framed area should not be routed and no components should be placed there.

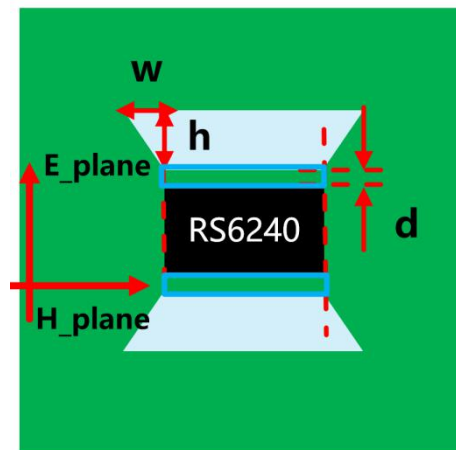
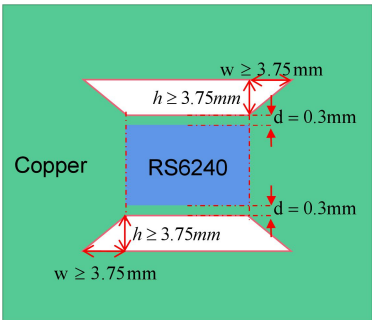


Figure 8-1 PCB Type 3 Floor Layout

Table 8-1 Reference design for mmWave radar PCB floor

Chip Type	RS6240
PCB Type 1	 No requirement for the length of the PCB along the long side of the chip PCB size of chip short edge: 10mm
PCB Type 2	 There is no requirement for the PCB size of the long side of the chip In cases where the short side of the chip PCB is greater than 16mm, a trapezoidal window shall be added to the PCB floor. The distance from the window to the edge of the chip shall be $d=0.3\text{mm}$, and the length of the parallel short side of the trapezoid shall be consistent with the length of the chip's long side $h \geq 3.75\text{mm}$, $w \geq 3.75\text{mm}$

8.2 Space Clearance

mmWave radar needs to be installed in a suitable environment to avoid the impact of other factors in the near-field environment on radar performance. Therefore, a good clearance area is required within the near-field region of the radar. The ideal clearance area should extend beyond the area **2.5mm around the radar**, and within the range of elevation $(180^\circ - \text{FOV})/2$, **good clearance is required to ensure the radiation coverage performance of the radar.**

- When arranging PCB components, the aforementioned constraints should be adhered to.
- When PCB space is limited, the requirements for the H direction can be relaxed. Components can be placed closely adjacent to RS6240, but **their height must not exceed that of the radar chip.**

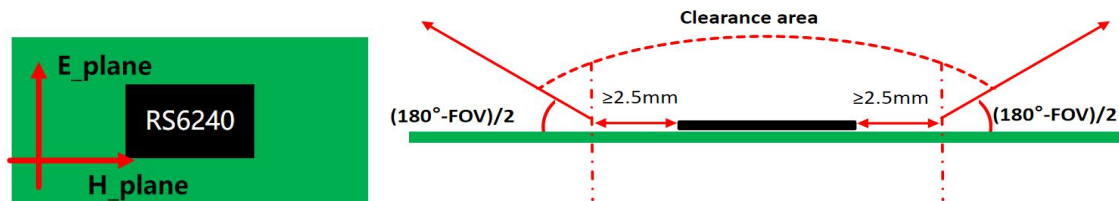


Figure 8-2 Space Clearance Requirements

8.3 Chip Placement Orientation

Due to the characteristics of the chip antenna, the radiation intensity in different directions may vary slightly. When no antenna cover is applied, the FOV (Field of View) in the E-plane is approximately 130 degrees, and in the H-plane, it is about 110 degrees. When a flat antenna cover is applied, the FOV in the E-plane is about 120 degrees, and in the H-plane, it is about 100 degrees. Therefore, if the usage scenario involves side mounting, it is recommended to place the chip vertically and use the E-plane to detect horizontal directions. As shown in the figure below.

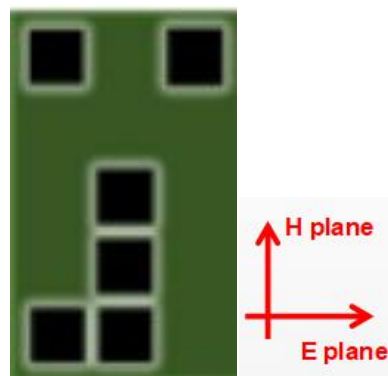


Figure 8-3 Chip Placement Orientation

8.4 Device Layout and Routing

Based on different application requirements, it can be divided into PCB single-sided layout and PCB double-sided layout.

PCB single-sided layout is suitable for applications where radar angle and pattern requirements are not stringent. Note that it is recommended not to use DCDC circuits when the board area is less than 30mm*30mm.

The PCB double-sided layout requires the radar chip to be placed on one side exclusively, which is suitable for applications demanding high precision in radar angle and directional pattern.

It is recommended to adopt a four-layer board design for the radar hardware application solution, and the following layout design guidelines will be based on a four-layer board.

8.4.1 PCB Single-sided Layout Requirements

- For the top-level device layout on one side, the chip should be placed in the middle of the board in a symmetrical position. Capacitors such as VRF, V13, VCC, VDDIO, V18, and VDIG must be placed closely adjacent to the chip and their corresponding signal pins, with a spacing of less than 2mm.
- The area outside the top-level E-direction chip is not routed to ensure the continuity and integrity of GND.

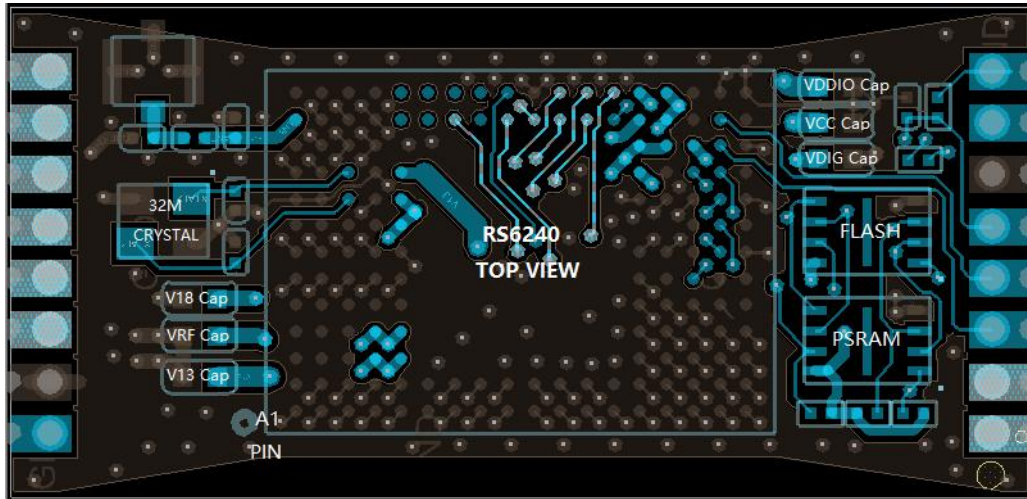


Figure 8-4 Top layer of PCB single-sided layout (radar chip side)

- The second layer of the single-sided layout is fully covered with GND, as shown below:

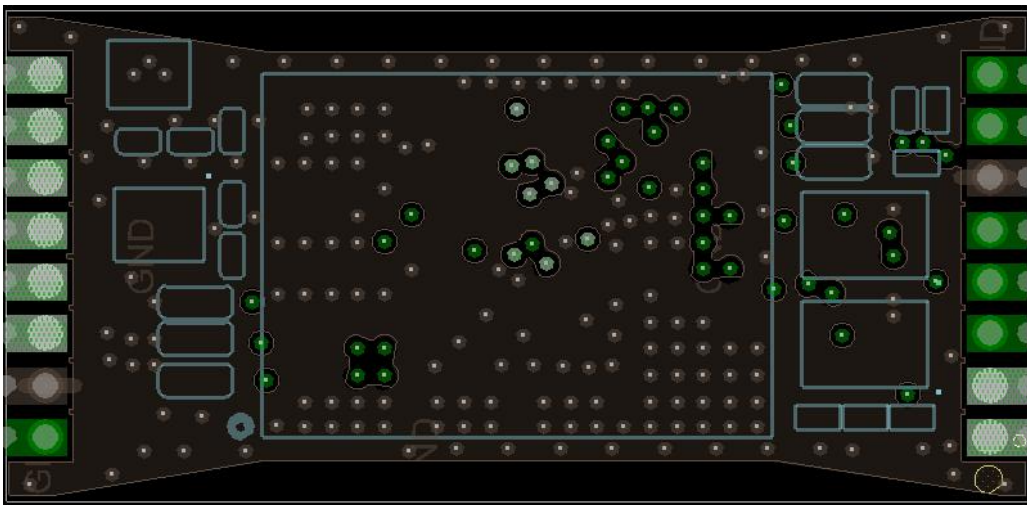


Figure 8-5 PCB single-sided layout, second layer with GND

- The power and signal traces are mainly distributed on the top layer, the third layer, and the bottom layer.
- Regarding the requirements for power line widths, the line widths of VCC and V13 must be greater than 0.5mm, while those of VDDIO, V18, and VRF must be greater than 0.25mm.
- V13 and V18 are power-sensitive, and their traces should not overlap with signal lines and VDDIO on adjacent layers.

8.4.2 PCB Double-sided Layout Requirements

TBD

8.4.3 DCDC Layout

- Schematic layout of DCDC matching circuit (corresponding to the [reference schematic diagram mentioned above](#)):

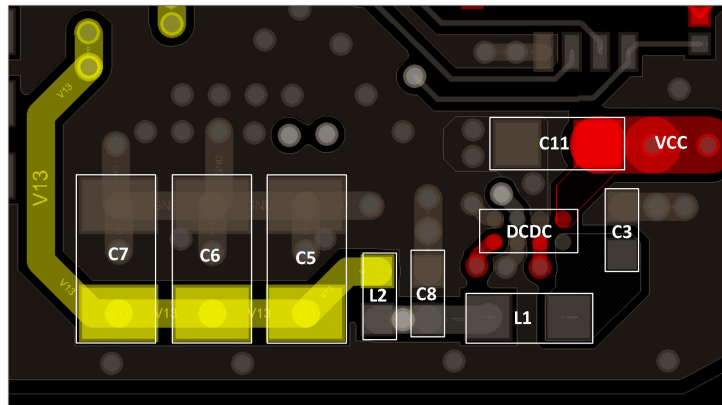


Figure 8-6 DCDC Layout

- DCDC circuit Layout requirements:
 - The DCDC matching circuit is laid out according to the layout shown in the figure above.
 - The surrounding GND of the DCDC must be complete, and the surrounding GND has been drilled.
 - [Recommendation] The DCDC and matching circuits should not be placed on the same layer as the radar chip, with a layout spacing of at least 1mm. The spacing area should have a complete and continuous GND.
 - When the DCDC and matching circuit are on the same layer as the radar chip, they must be spaced at least 5mm apart during layout, and the spacing area should have a complete and continuous GND.
 - Regardless of whether the DCDC is on the same layer as the radar chip, it must be kept away from the V13T and V13R pins of the chip. In particular, inductors, magnetic beads, and the DCDC chip must be kept at least 15mm away on the same layer, and at least 12mm away on different layers.

8.4.4 32M clock Layout

- Top-level layout schematic diagram of 32M crystal oscillator clock:

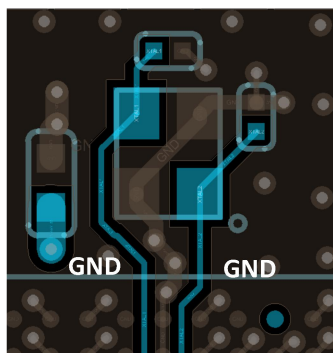


Figure 8-7 Clock Layout

- Layout requirements for 32M crystal oscillator clock:
 - The 32M crystal oscillator should be isolated from power supplies V13 and V18 using GND, and the wiring should be as short as possible.
 - Lay GND on both sides of the X1 and X2 signal traces to reduce unnecessary signal coupling.
 - The crystal oscillator GND via connects to the second layer GND.

8.4.5 Antenna Layout

- Schematic diagram of BLE antenna top-level layout:

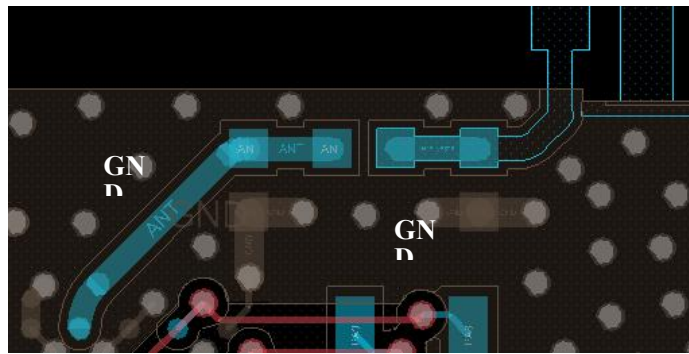


Figure 8-8 BLE Antenna Layout

- Precautions for BLE antenna layout:
- The antenna wiring should be as short as possible.
 - The impedance of antenna wiring must be 50Ω , and impedance control should refer to the third-layer GND.
 - Lay GND near the antenna wiring.
 - The GND vias on both sides of the antenna are symmetrical.
 - The matching circuit needs to be modified according to the type of antenna selected.

8.4.6 External SPI Layout

If users utilize the SPI interface for communication with an external host, the following points should be noted:

- The trace impedance is controlled at 50 ohms (plus or minus 10%) per end.
- The requirement for equal length between the clock and data is less than 5mm.
- The length of the wiring is less than 10cm.
- The spacing between signal lines should be at least twice the line width.

RS6240

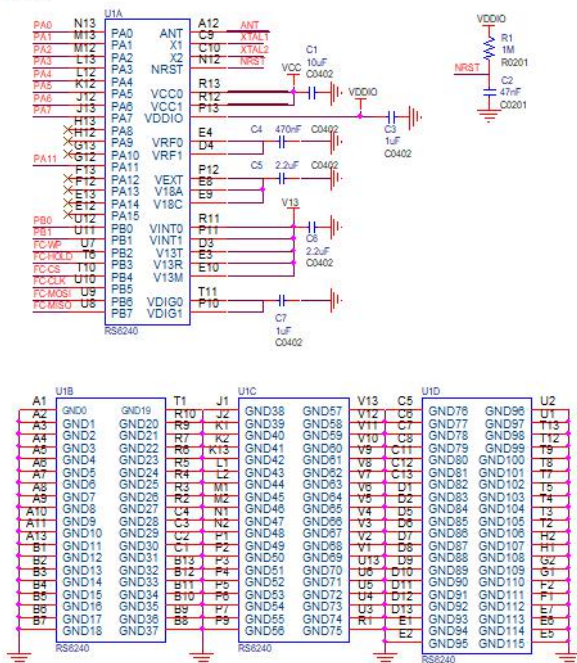


Figure 8-9 Schematic Diagram of P2512 Module

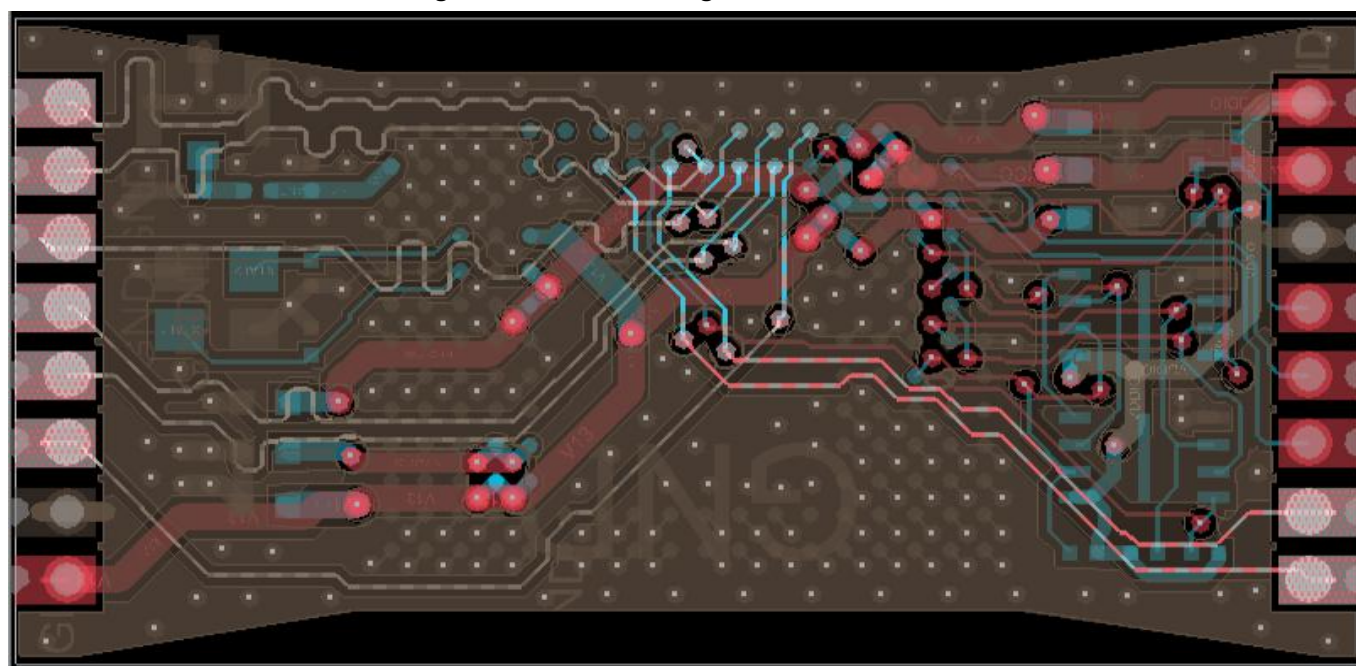
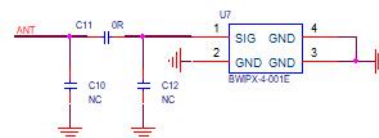
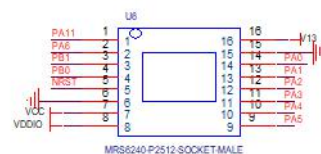


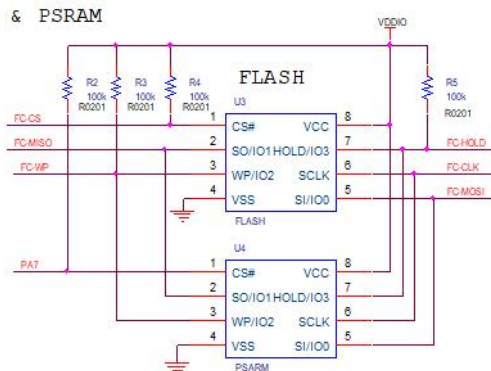
Figure 8-10 Overview



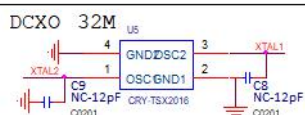
INTERFACE



SPI NOR & PSRAM



CLOCK



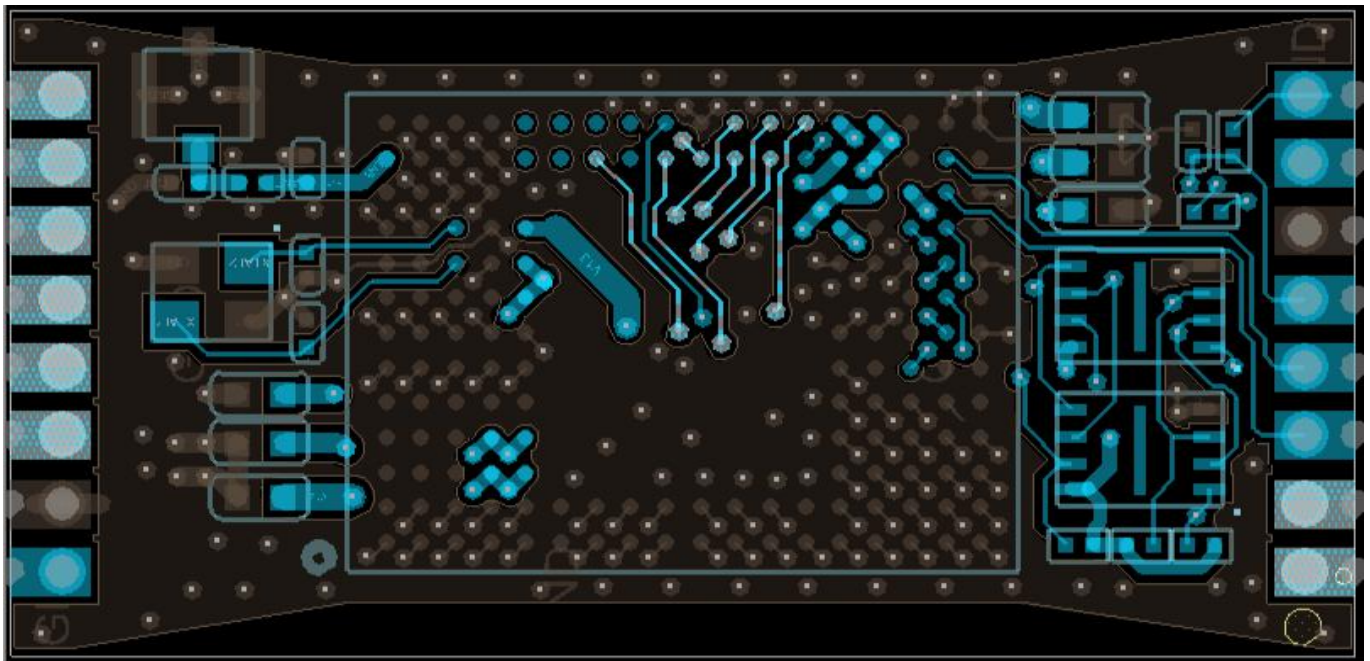


Figure 8-11 Top Layer

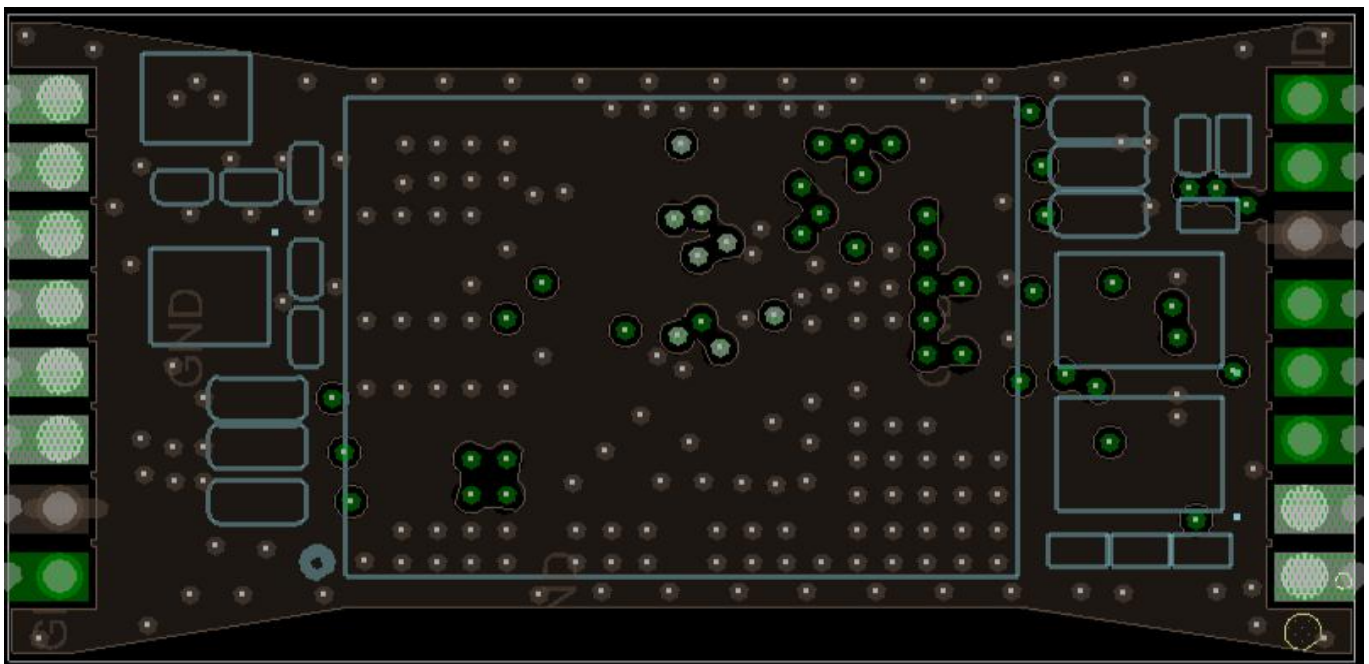


Figure 8-12 Second Layer

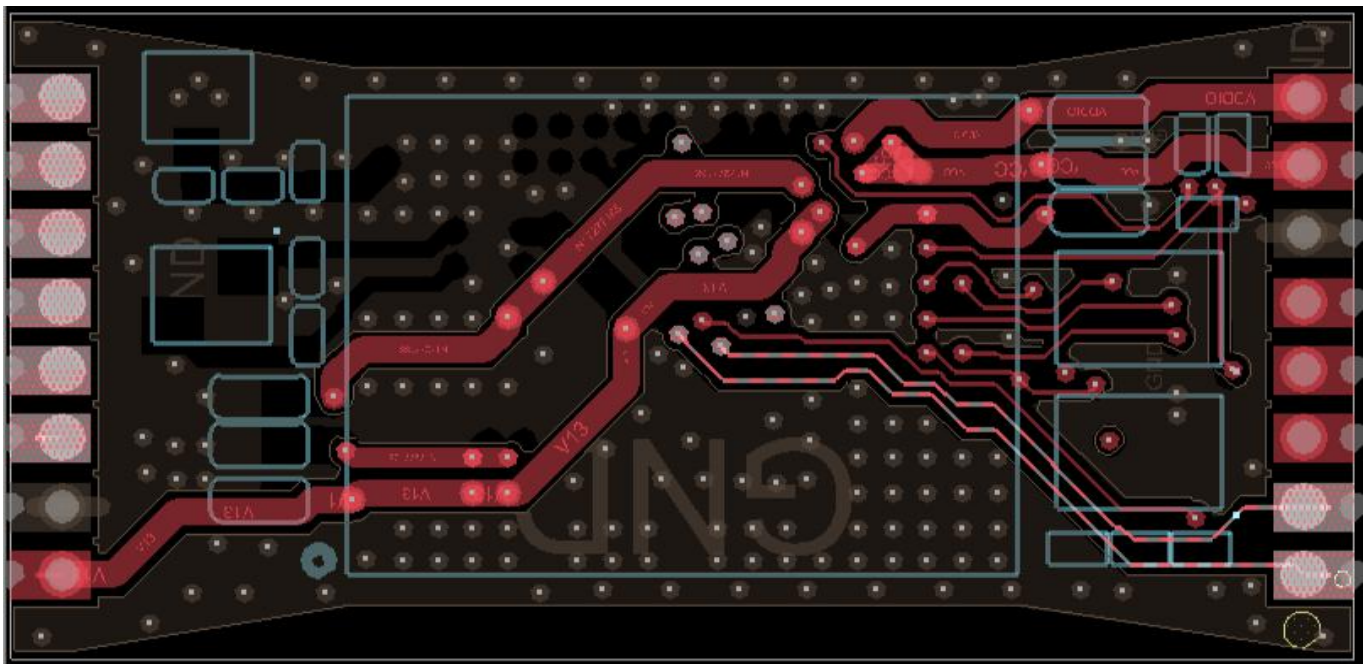


Figure 8-13 Third Layer

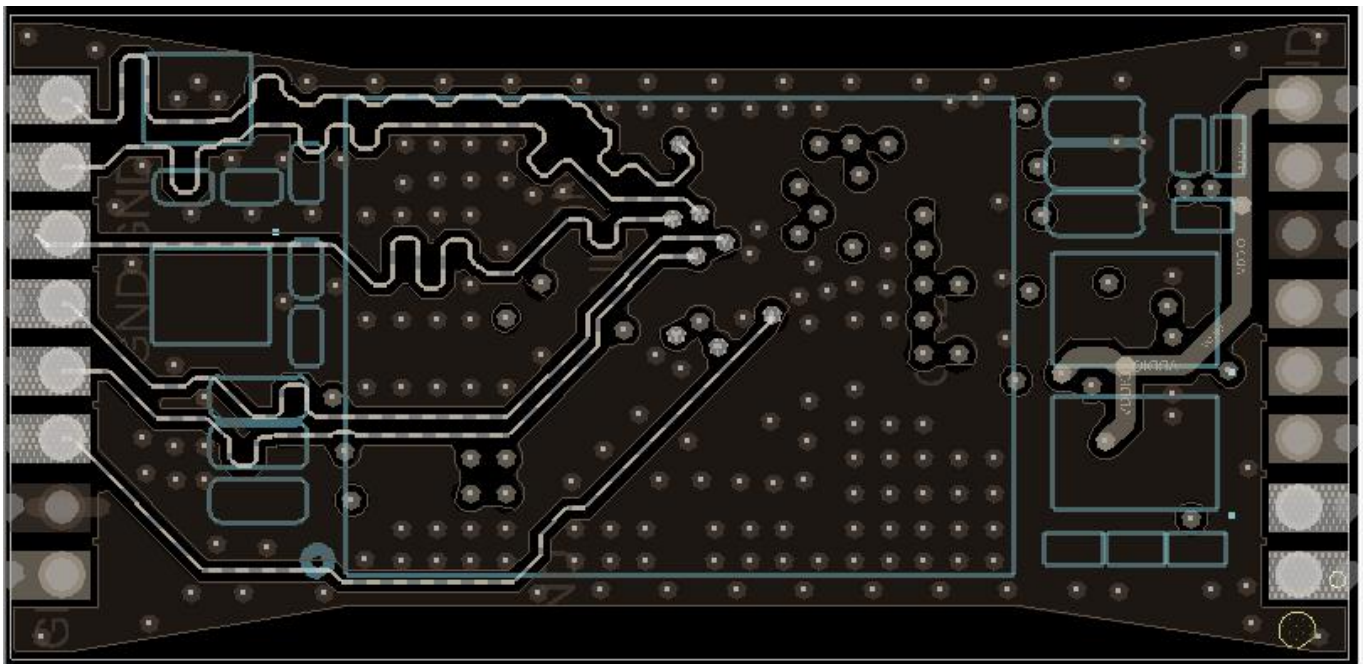


Figure 8-14 Bottom Layer

9 PCB Performance Analysis

9.1 Noise Floor Measurement

Poor PCB design can introduce excessive noise into the radar system, thereby affecting ranging performance. Below, we will introduce a method for measuring noise, so that users can evaluate the quality of PCB design.

The test method requires the use of PA0 (UART1-TX), PB0 (UART0-TX), and PB1 (UART0-RX). Therefore, during PCB design, it is necessary to ensure that these three signals can be brought out.

Test preparation:

- Upper computer tool "RadarAnalysisTool"
- PCB to be tested (with the dedicated test firmware for programming, "radar_analysis_rs6240_p2512.img")
- Serial port adapter board (connecting UART0-TX, UART0-RX, UART1-TX of PCB)
- Test environment: Ensure that only the radar under test is operating in the surrounding area, as the operation of other radars may cause interference.

Note: The host computer tool and dedicated test firmware can be obtained from SDK\Software_Kit\04_image.

9.1.1 Usage of The Host Computer

Step 1: Power on the PCB, connect the aforementioned two serial ports to the PC, and open the host computer tool.

- 1) Click on "Platform".
- 2) Select the hardware platform "MRS6240-2512".
- 3) Connect to the serial port, select the COM port corresponding to UART0 for the control port, with a baud rate of 115200, and select the COM port corresponding to UART1 for the data port, with a baud rate of 1000000.
- 4) Click "Connect Device".

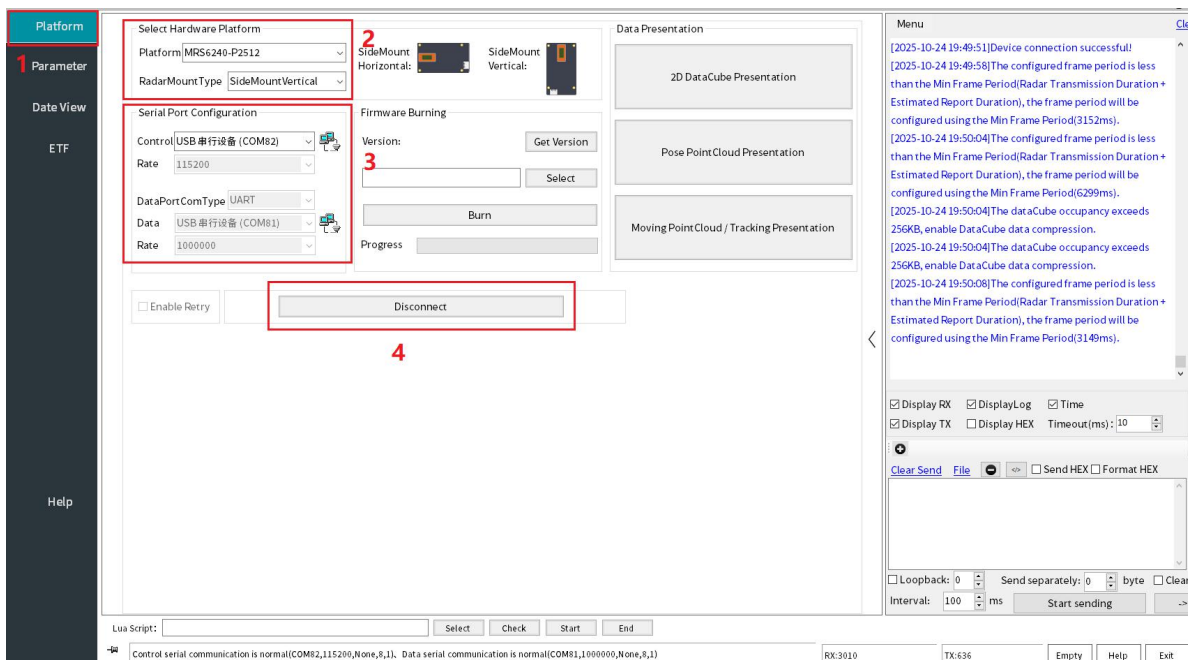


Figure 9-1 PCB Performance Test Step 1

Step 2:

- 1) Click on "Parameter".
- 2) Set the following parameter in order:Antenna Mode(2T4R),Start freq(58),Max distance(10),Max speed(10),Range resolution(4),Speed resolution(64),frame structure(2DFFT),Interval chip(1),Frame Num(0),Frame Period(3149)
- 3) click "Start".

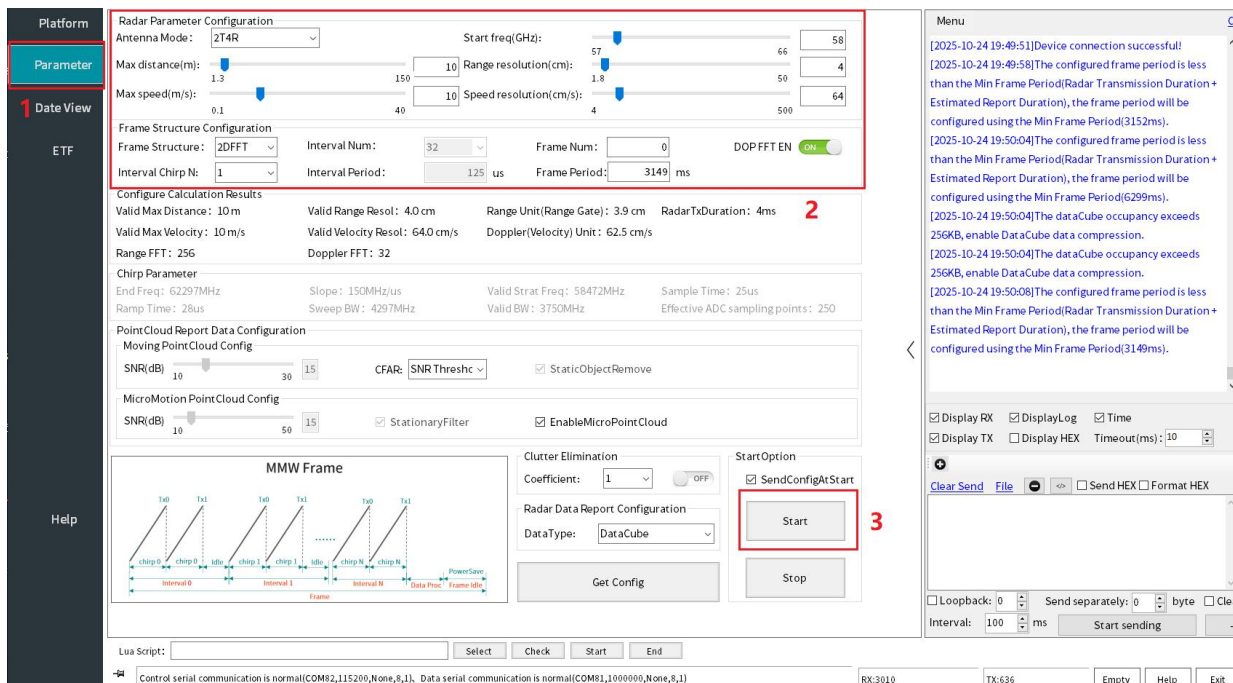


Figure 9-2 PCB Performance Test Step 2

Step three:

- 1) Click on "Data View" on the page.
- 2) Click the "Display Mode" button and select "One Place Grid".
- 3) Check the "Option" and uncheck "Max Value" in the pop-up menu.
- 4) Finally, set the "static removal" parameter to "10".

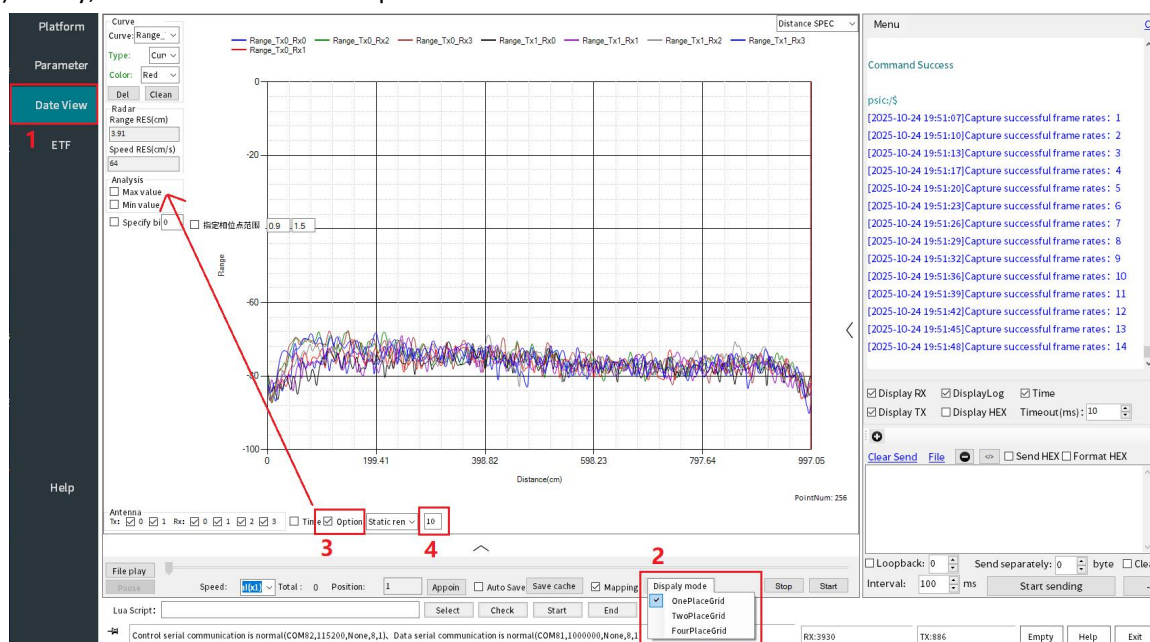


Figure 9-3 PCB Performance Test Step 3

9.1.2 Criteria for Judgment

Under the condition of good hardware design, the measured noise characteristics are as follows:

- The amplitude is below -72dB within 0~2m, below -76dB within 2~4m, and below -80dB within 4~10m.
- The extreme amplitude difference of different polylines at the same distance is less than 10dB.

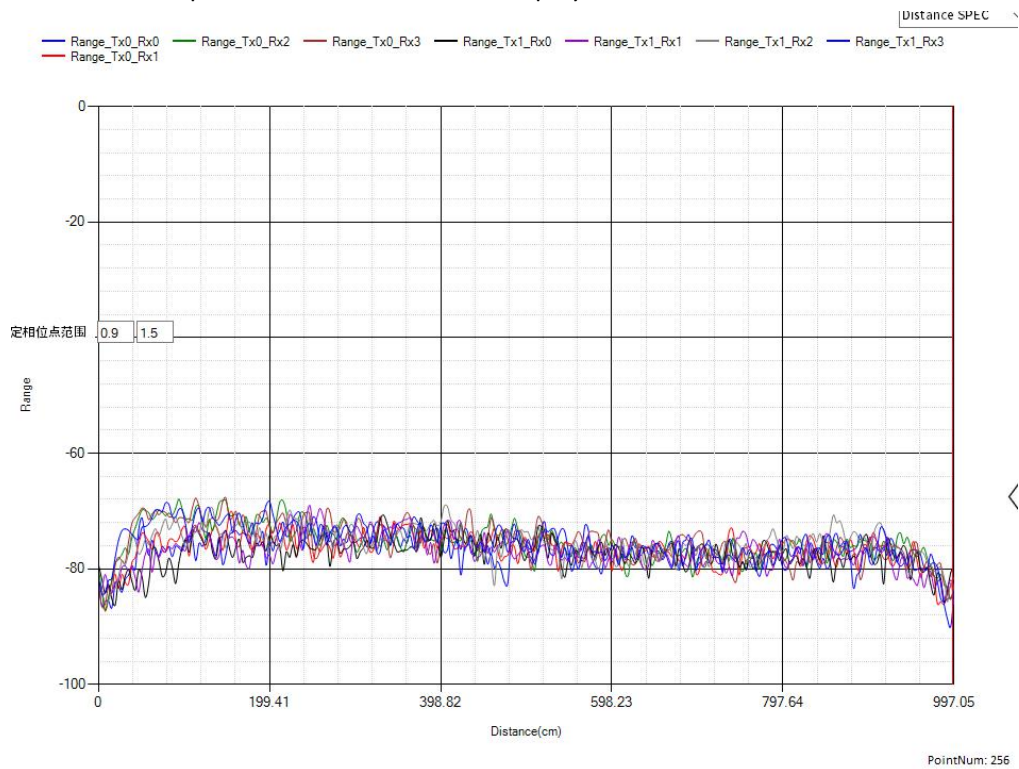


Figure 9-4 Normal graph

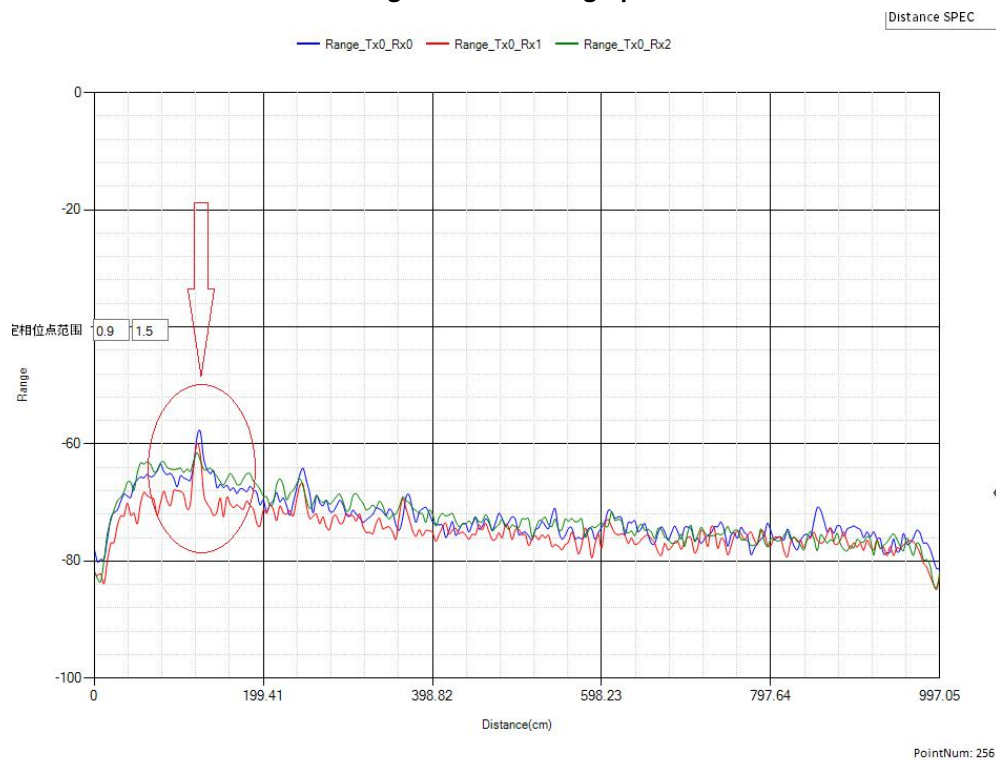


Figure 9-5 Abnormal Pattern (Case of RS6130 Being Interfered by DCDC Noise)

9.2 DCDC Output Performance Measurement

If the user's solution utilizes a DCDC, its output performance needs to be measured to ensure it meets the radar V13 power input requirements.

- When the radar is operating, it periodically emits mmWave to detect the environment. Therefore, the V13 power supply will exhibit periodic power extraction behavior. At the moment of emission initiation, a short-term voltage drop will inevitably occur in V13. However, the external circuit design should minimize the DCR on the DCDC output to the chip V13 pin path, ensuring that the drop amplitude is less than 100mV, and the voltage after the drop is greater than 1.2V.
- The switching noise of the DCDC can couple to the V13 network, forming power supply ripple (excluding the voltage drop during the aforementioned launch transient). Excessively high ripple can interfere with radar RF. Therefore, when designing the external circuit, appropriate capacitors should be placed in the V13 power supply network to ensure that the power supply ripple V_{pp} measured by V13 is less than 50mV.

9.3 PCB Performance Verification Scheme

Table 9-1 PCB Performance Verification Scheme

Project	Verification Standard
Power Consumption	Using the actual measurement data from the development board under the same configuration as the comparison standard, the main focus is on the base current and average current
Power Circuit Quality	The drop amplitude of VCC is less than 100mV
	The drop amplitude of V13 is less than 50mV, with a maximum of no more than 100mV, and the voltage after drop is greater than 1.2V
	The power supply ripple V_{pp} of V13 is less than 50mV
Noise Floor	The extreme amplitude difference of different polylines at the same distance is less than 10dB
	The amplitude is below -72dB for 0~2m, below -76dB for 2~4m, and below -80dB for 4~10m
Angles Performance	Program the DEMO firmware, perform distance measurement/angle measurement, and compare with the development board

9.4 Q & A

Table 9-2 Q & A

Project	Problem	Solution
Power Circuit Quality	VCC drop is too large	Check the power supply capacity of the circuit
	V13 ripple is large	Check whether the output capacitance of V13 is sufficient
	V13 has a large transient drop	According to the BOM table in the PCB manual, check whether the DCR of the inductor/magnetic bead is too large, and whether the output capacitor is sufficient
Noise Floor	Excessive background noise	Remove the DCDC and use internal LDO for power supply to confirm whether the issue is caused by the DCDC.
Angles Performance	If the performance does not meet the requirements	Re-check power quality and background noise
		Check whether the PCB floor design meets the requirements

10 Design Checklist

Users are required to conduct self-inspection against the following check items both before and after designing the schematic diagram/PCB.

10.1 Schematic Checklist

Table 10-1 Schematic Checklist

External Connection Signal	Communication Interface	Supports UART/I2C/SPI communication. After selecting the interface, please refer to Section 7.1 "Precautions" and checks. If UART/SPI is chosen, it is necessary to ensure that the corresponding signals on the main control terminal (pin RX (PB1) for UART0, and pins CS (PA0) and CLK (PA1) for SPI0) are configured as pull-up or pull-down when the main control is in hibernation. Otherwise, the radar may experience abnormal leakage. If the main control cannot provide the pull-up/pull-down configuration during hibernation, it is necessary to configure pull-up/pull-down resistors on the periphery.
	Control Signal	Regardless of which communication interface is chosen, it is recommended that the master control terminal allocate IOs to control the RESET and PB0 signals of the radar. When it is necessary to update the radar firmware online, the radar chip can be reset via the RESET signal. If PB0 is detected as being pulled low during the reset process, the radar will enter firmware programming mode, allowing the main controller to update the radar firmware.
	Trigger Signal	PA6/PA11 is preferred as the signal for triggering the host.
	Test Point	Reserve PB0, PB1, and PA0 for radar performance analysis. Our company provides radar analysis firmware and host computer tools. The radar needs to interact with the host computer through UART1-TX (PA0)/UART0-TX (PB0)/UART0-RX (PB1).
Power Supply Design	VCC	Supports 1.7~5.5V, with 3.3V recommended. When the radar is operating, it periodically emits mmWave to detect targets, resulting in periodic fluctuations on the VCC. After adding a filter capacitor, the fluctuation is generally within 100mV. If there are voltage-sensitive devices on the periphery, such as a speaker amplifier, do not share the VCC with the radar.
	VDDIO	Supports 1.7~3.6V, with 3.3V being recommended. If VCC is not greater than 3.3V, VDDIO can be connected to VCC.
		The flash is powered by VDDIO. If a VDDIO voltage below 3.3V is chosen, it is important to select a wide-voltage or low-voltage type of flash.
	V13	V13 can be powered by an internal LDO. If low power consumption is required, it should be switched to an external DCDC for power supply. The radar RF is sensitive to external DCDC switching noise. Please refer to the device description in Section 7.2 to check whether the device selection is correct.
Flash Selection	Other Designs	The remaining power supplies can be designed based on the schematic diagram referenced in Section 7.2 . Please ensure that the capacitance values of the peripheral capacitors are consistent with those shown in the reference schematic diagram.
	Model Selection	Refer to the "Key Material Support List" to select the flash model. If a model not listed in the support list is required, you need to add the parameters yourself and conduct

		verification testing.		
	Capacity Selection	If the internal OTA update function of the radar is not used "Internal OTA" refers to the process of first downloading firmware to the backup area stored in the radar, and then allowing the radar to complete the firmware update on its own.	Development-free solution	Choose a flash memory with a capacity of no less than 512KB
			SDK solution, i.e. customizing radar firmware	If not using the XIP scheme, the flash capacity should be chosen to be no less than 512KB
				If the XIP scheme is used, the flash capacity should be larger than the size of the compiled firmware
		If the internal OTA update function of the radar is used	Then double the capacity for the scenarios listed under "Do not use OTA" to provide redundant space for storing new firmware	
Attention	If using WINBOND's flash, it is necessary to reserve pull-up resistors for MOSI/MISO to reduce sleep leakage at the flash end.			
Other	For component selection based on the schematic diagram, please refer to the instructions in Section 7.2 "Reference Schematic" to check whether the selection is correct.			
	The capacitance around the chip must be designed strictly in accordance with the capacitance values specified in Section 7.2 "Reference Schematic" .			

10.2 PCB Layout Checklist

Table 10-2 PCB Layout Checklist

Floor Layout	The GND layout around the radar chip has a significant impact on antenna performance. Please refer to Section 8.1 “Floor Layout” for design, especially for solutions with cross-cutting needs/high angle measurement requirements, where the guidelines must be strictly followed	
	The PCB structure surrounding the chip should be as symmetrical as possible, especially for solutions with angle measurement/cross-cutting requirements. Asymmetrical structures will produce asymmetrical directional patterns.	
Chip Placement Method	In the top-loading scenario, there are no specific requirements for chip placement.	
Device Layout	No DCDC Solution	Refer to Section 8.5 “Single-sided Layout Example - P2512 Module” , where devices are distributed on both sides of the short edge of the chip, with no devices placed on both sides of the long edge.
	Equipped with DCDC Solution	Please refer to Section 8.4.3 “DCDC Layout”
	The capacitors around the chip are placed close to the corresponding pins, with a spacing of 2mm.	
Lamination	At least four layers of board	Top layer: Place the radar chip, and prioritize placing other components on the bottom layer. Do not route wires around the radar chip. A complete GND is more conducive to improving angle measurement/cross-section performance.
		Layer 2: Set as a complete GND.
		Layer 3: Routing as needed.
		Bottom layer: Place the remaining components.
Routing	VCC/V13 will experience a peak current of 100~200mA. The preferred wiring should be larger than 0.5mm, and at least larger than 0.3mm.	
	V13 and V18 are power-sensitive, and their traces should not overlap with signal lines and VDDIO on adjacent layers.	
	If SPI is used as the communication interface, the external SPI Layout needs to be routed according to Section 8.4.6 .	
	The routing width of the VCC input for the DCDC needs to be greater than 0.5mm, and the filtering capacitor for VCC needs to be close to the VCC pin of the DCDC.	

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